

Confidence Intervals

1. Let $X_1, \dots, X_n | \boldsymbol{\theta} \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ where $\boldsymbol{\theta} = (\mu, \sigma^2)$ unknown. Obtain a $\gamma \times 100\%$ confidence interval for σ^2 . *Be careful about inequality rules!* Is your interval exact?

We begin by finding a pivot! That is, we need a random variable that contains both the data $\mathbf{X}$ and σ^2 but whose distribution is independent of σ^2 . One such random variable is

$$U = \frac{n}{\sigma^2} \hat{\sigma}_{MLE}^2 \sim \chi_{n-1}^2 \quad (\text{Thm. 8.3.1B})$$

Now, we need to find values a and b such that $\gamma = P(a \leq U \leq b)$. Letting $a = F_U^{-1}(\frac{1-\gamma}{2})$ and $b = F_U^{-1}(\frac{1+\gamma}{2})$ where F_U^{-1} is the inverse CDF of a χ_{n-1}^2 distribution, we have found our a and b . Thus,

$$\begin{aligned} \gamma &= P(a \leq U \leq b) \\ &= P\left(a \leq \frac{n}{\sigma^2} \hat{\sigma}_{MLE}^2 \leq b\right) \\ &= P\left(\frac{a \hat{\sigma}_{MLE}^2}{n} \leq \frac{1}{\sigma^2} \leq \frac{b \hat{\sigma}_{MLE}^2}{n}\right) \\ &= P\left(\frac{n}{a \hat{\sigma}_{MLE}^2} \geq \sigma^2 \geq \frac{n}{b \hat{\sigma}_{MLE}^2}\right) \end{aligned}$$

Thus, an exact γ coefficient CI for σ^2 is

$$\left(\frac{n}{F_U^{-1}(\frac{1+\gamma}{2})} \hat{\sigma}_{MLE}^2, \frac{n}{F_U^{-1}(\frac{1-\gamma}{2})} \hat{\sigma}_{MLE}^2 \right)$$

2. Let $X_1, \dots, X_n | \boldsymbol{\theta} \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ where $\boldsymbol{\theta} = (\mu, \sigma^2)$ unknown. We want to construct an exact one-sided γ -coefficient confidence interval for μ of the form: $(A(\mathbf{X}), \infty)$.

- (a) Clearly state the probability statement about $A(\mathbf{X})$ involving μ that this interval estimator should satisfy. i.e. $\Pr(\dots) = \dots$

We want to find $A(\mathbf{X})$ such that $\Pr(A(\mathbf{X}) \leq \mu) = \gamma$.

- (b) Now find your confidence interval!

Since both the mean and variance are unknown, consider the pivot $U = \frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}$. We might begin by finding some value a such that

$$\gamma = P(a \leq U)$$

But if we continue down this path, we notice that inverting results in the following:

$$\gamma = P\left(a \leq \frac{\sqrt{n}(\bar{X} - \mu)}{S}\right) = P\left(\frac{aS}{\sqrt{n}} \leq \bar{X} - \mu\right) = P\left(\mu \leq \bar{X} - \frac{aS}{\sqrt{n}}\right)$$

However, this is the wrong-direction! This results in an interval estimator of the form $(-\infty, B(\mathbf{X}))$. So instead, we want the value a such that

$$\gamma = P(U \leq a) \Rightarrow a = T_{n-1}^{-1}(\gamma),$$

where T_{n-1}^{-1} is the inverse CDF of the t_{n-1} distribution. Then we can invert the pivot as follows:

$$\gamma = P(U \leq a) = P\left(\frac{\sqrt{n}(\bar{X} - \mu)}{S} \leq a\right) = P\left(\mu \geq \bar{X} - \frac{aS}{\sqrt{n}}\right)$$

So an exact one-sided $100 \times \gamma\%$ CI for μ of the desired form is $(\bar{X} - T_{n-1}^{-1}(\gamma) \frac{S}{\sqrt{n}}, \infty)$